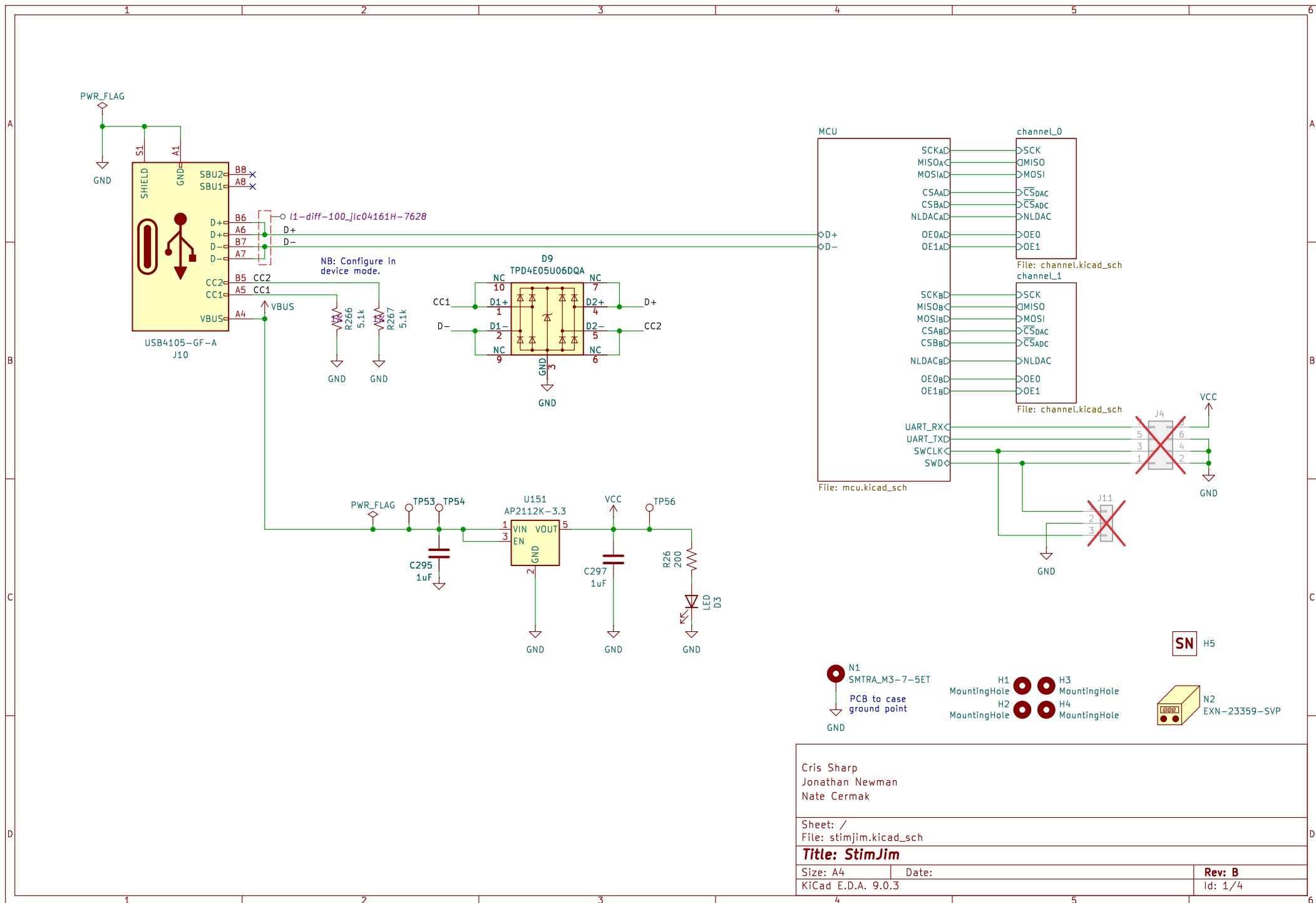
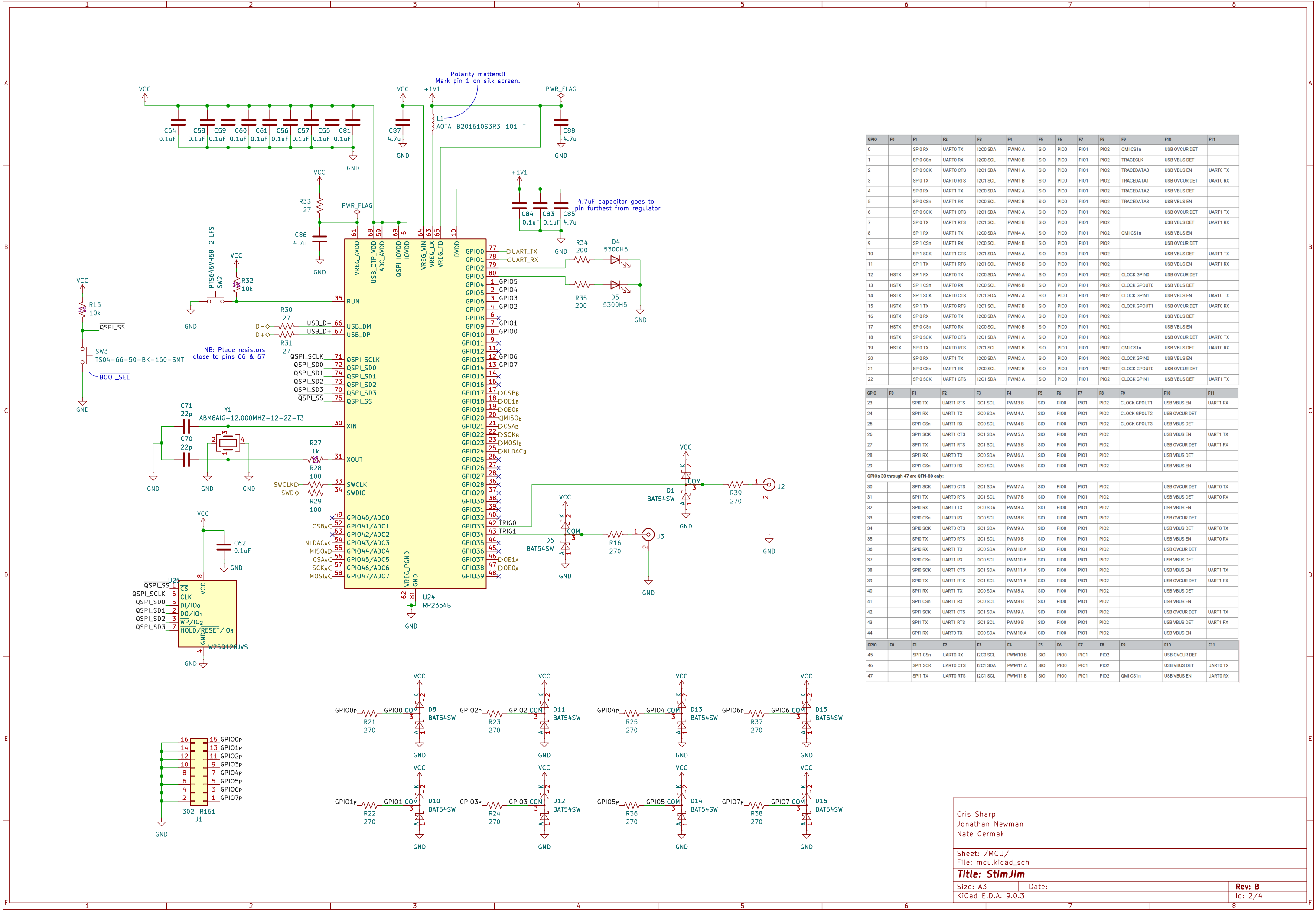






GPIO	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
0		SPIO RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PI00	PI01	PI02	QMI CS1n	USB OVCUR DET	
1		SPIO CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PI00	PI01	PI02	TRACEDATA0	USB VBUS DET	
2		SPIO SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PI00	PI01	PI02	TRACEDATA0	USB VBUS EN	UART0 TX
3		SPIO TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PI00	PI01	PI02	TRACEDATA1	USB OVCUR DET	UART0 RX
4		SPIO RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PI00	PI01	PI02	TRACEDATA2	USB VBUS DET	
5		SPIO CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PI00	PI01	PI02	TRACEDATA3	USB VBUS EN	
6		SPIO SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PI00	PI01	PI02		USB OVCUR DET	UART1 TX
7		SPIO TX	UART1 RTS	I2C1 SCL	PWM3 B	SIO	PI00	PI01	PI02		USB VBUS DET	UART1 RX
8		SPI1 RX	UART1 TX	I2C0 SDA	PWM4 A	SIO	PI00	PI01	PI02	QMI CS1n	USB VBUS EN	
9		SPI1 CSn	UART1 RX	I2C0 SCL	PWM4 B	SIO	PI00	PI01	PI02		USB OVCUR DET	
10		SPI1 SCK	UART1 CTS	I2C1 SDA	PWM5 A	SIO	PI00	PI01	PI02		USB VBUS DET	UART1 TX
11		SPI1 TX	UART1 RTS	I2C1 SCL	PWM5 B	SIO	PI00	PI01	PI02		USB VBUS EN	UART1 RX
12	HSTX	SPI1 RX	UART0 TX	I2C0 SDA	PWM6 A	SIO	PI00	PI01	PI02	CLOCK GPIN0	USB OVCUR DET	
13	HSTX	SPI1 CSn	UART0 RX	I2C0 SCL	PWM6 B	SIO	PI00	PI01	PI02	CLOCK GPOUT0	USB VBUS DET	
14	HSTX	SPI1 SCK	UART0 CTS	I2C1 SDA	PWM7 A	SIO	PI00	PI01	PI02	CLOCK GPIN1	USB VBUS EN	UART0 TX
15	HSTX	SPI1 TX	UART0 RTS	I2C1 SCL	PWM7 B	SIO	PI00	PI01	PI02	CLOCK GPOUT1	USB OVCUR DET	UART0 RX
16	HSTX	SPIO RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PI00	PI01	PI02		USB VBUS DET	
17	HSTX	SPIO CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PI00	PI01	PI02		USB VBUS EN	
18	HSTX	SPIO SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PI00	PI01	PI02		USB OVCUR DET	UART0 TX
19	HSTX	SPIO TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PI00	PI01	PI02	QMI CS1n	USB VBUS DET	UART0 RX
20		SPIO RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PI00	PI01	PI02	CLOCK GPIN0	USB VBUS EN	
21		SPIO CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PI00	PI01	PI02	CLOCK GPOUT0	USB OVCUR DET	
22		SPIO SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PI00	PI01	PI02	CLOCK GPIN1	USB VBUS DET	UART1 TX

GPIO	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
23		SPIO TX	UART1 RTS	I2C1 SCL	PWM3 B	SIO	PI00	PI01	PI02	CLOCK GPOUT1	USB VBUS EN	UART1 RX
24	SPI1 RX	UART1 TX	I2C0 SDA	PWM4 A	SIO	PI00	PI01	PI02	CLOCK GPOUT2	USB OVCUR DET		
25	SPI1 CSn	UART1 RX	I2C0 SCL	PWM4 B	SIO	PI00	PI01	PI02	CLOCK GPOUT3	USB VBUS DET		
26		SPI1 SCK	UART1 CTS	I2C1 SDA	PWM5 A	SIO	PI00	PI01	PI02		USB VBUS EN	UART1 TX
27		SPI1 TX	UART1 RTS	I2C1 SCL	PWM5 B	SIO	PI00	PI01	PI02		USB OVCUR DET	UART1 RX
28		SPI1 RX	UART0 TX	I2C0 SDA	PWM6 A	SIO	PI00	PI01	PI02		USB VBUS DET	
29		SPI1 CSn	UART0 RX	I2C0 SCL	PWM6 B	SIO	PI00	PI01	PI02		USB VBUS EN	

GPIOs 30 through 47 are QFN-80 only:

30		SPI1 SCK	UART0 CTS	I2C1 SDA	PWM7 A	SIO	PI00	PI01	PI02		USB OVCUR DET	UART0 TX
31		SPI1 TX	UART0 RTS	I2C1 SCL	PWM7 B	SIO	PI00	PI01	PI02		USB VBUS DET	UART0 RX
32		SPIO RX	UART0 TX	I2C0 SDA	PWM8 A	SIO	PI00	PI01	PI02		USB VBUS EN	
33		SPIO CSn	UART0 RX	I2C0 SCL	PWM8 B	SIO	PI00	PI01	PI02		USB OVCUR DET	
34		SPIO SCK	UART0 CTS	I2C1 SDA	PWM9 A	SIO	PI00	PI01	PI02		USB VBUS DET	UART0 TX
35		SPIO TX	UART0 RTS	I2C1 SCL	PWM9 B	SIO	PI00	PI01	PI02		USB VBUS EN	UART0 RX
36		SPIO RX	UART1 TX	I2C0 SDA	PWM10 A	SIO	PI00	PI01	PI02		USB OVCUR DET	
37		SPIO CSn	UART1 RX	I2C0 SCL	PWM10 B	SIO	PI00	PI01	PI02		USB VBUS DET	
38		SPIO SCK	UART1 CTS	I2C1 SDA	PWM11 A	SIO	PI00	PI01	PI02		USB VBUS EN	UART1 TX
39		SPIO TX	UART1 RTS	I2C1 SCL	PWM11 B	SIO	PI00	PI01	PI02		USB OVCUR DET	UART1 RX
40		SPI1 RX	UART1 TX	I2C0 SDA	PWM8 A	SIO	PI00	PI01	PI02		USB VBUS DET	
41		SPI1 CSn	UART1 RX	I2C0 SCL	PWM8 B	SIO	PI00	PI01	PI02		USB VBUS EN	
42		SPI1 SCK	UART1 CTS	I2C1 SDA	PWM9 A	SIO	PI00	PI01	PI02		USB OVCUR DET	UART1 TX
43		SPI1 TX	UART1 RTS	I2C1 SCL	PWM9 B	SIO	PI00	PI01	PI02		USB VBUS DET	UART1 RX
44		SPI1 RX	UART0 TX	I2C0 SDA	PWM10 A	SIO	PI00	PI01	PI02		USB VBUS EN	

GPIO	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
45		SPI1 CSn	UART0 RX	I2C0 SCL	PWM10 B	SIO	PI00	PI01	PI02		USB OVCUR DET	
46		SPI1 SCK	UART0 CTS	I2C1 SDA	PWM11 A	SIO	PI00	PI01	PI02		USB VBUS DET	UART0 TX
47		SPI1 TX	UART0 RTS	I2C1 SCL	PWM11 B	SIO	PI00	PI01	PI02	QMI CS1n	USB VBUS EN	UART0 RX

Cris Sharp
Jonathan Newman
Nate Cermak

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File: mcu.kicad_sch

Title: **StimJim**

Size: A3

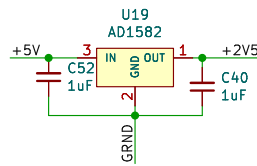
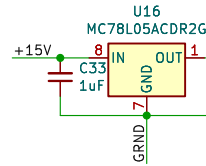
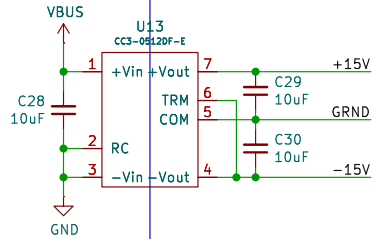
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Rev: **B**

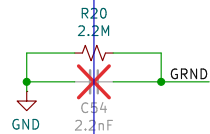
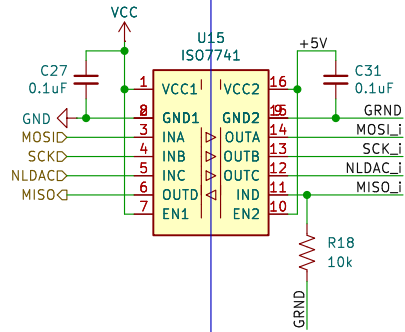
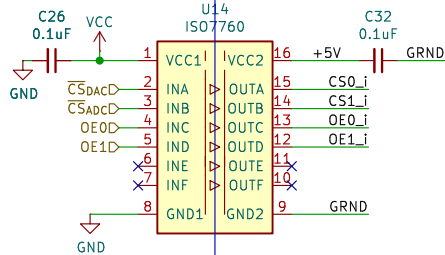
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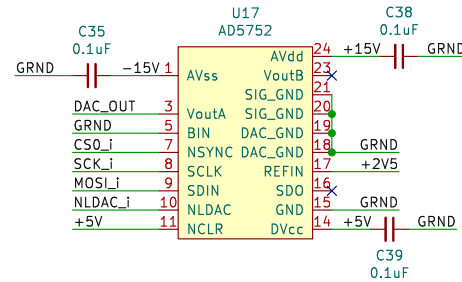
POWER RAILS
power rails are defined as local nets since they are galvanically isolated



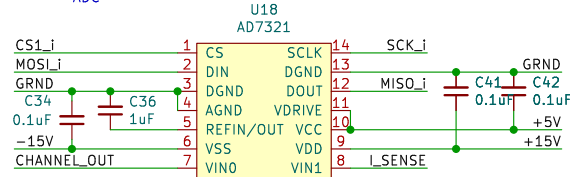
SIGNAL ISOLATION



DAC



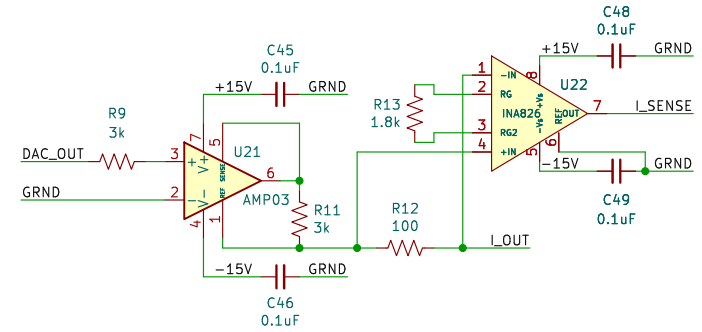
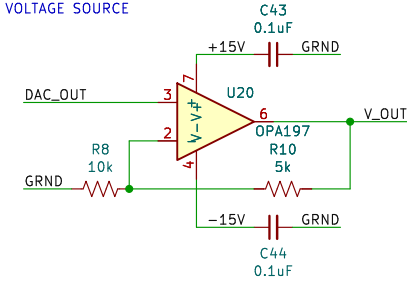
ADC



+15V ○ TP17 TestPoint
-15V ○ TP18 TestPoint
+5V ○ TP20 TestPoint
+2V5 ○ TP22 TestPoint
GND ○ TP31 TestPoint
GND ○ TP32 TestPoint
GND ○ TP33 TestPoint

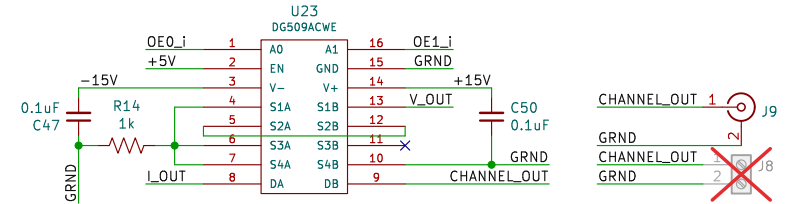
L_OUT ○ TP28 TestPoint
L_SENSE ○ TP30 TestPoint
DAC_OUT ○ TP24 TestPoint
V_OUT ○ TP26 TestPoint
MISO_i ○ TP34 TestPoint
MOSI_i ○ TP37 TestPoint
OE1_i ○ TP38 TestPoint
OE0_i ○ TP39 TestPoint
CS1_i ○ TP40 TestPoint
CS0_i ○ TP41 TestPoint

VOLTAGE SOURCE



Max current output (0ohm load) is $(+/-10V)/3k = +/- 3.33mA$.
Current resolution is $20V/2^{16}/3k = 0.1uA$
Max voltage output (infinite load) is $40/43 * 14.8V = +/- 13.76V$

gain is $R_{sense} * (1 + 49.4k/R_g) = 100 * (1 + 49.4k/1.8k) = 2.87 mV/uA$
Voltage output range is thus $+/- 3333 * 0.00287 = +/- 9.57V$
output needs to be within $+/- 10V$
current sensing resolution is $10000 mV / 2^{12} / (2.87mV/uA) = 0.85uA$



OE0	OE1	Output
0	0	VOLTAGE
1	0	CURRENT
0	1	High-Z
1	1	GND

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Sheet: /channel_0/
File: channel.kicad_sch

Title: StimJm

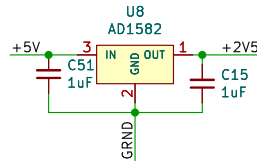
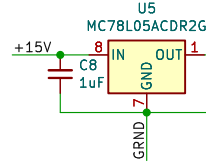
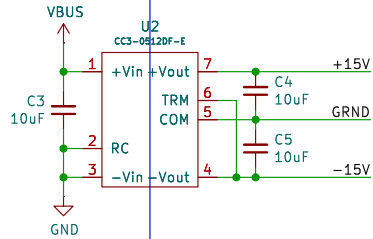
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Date:

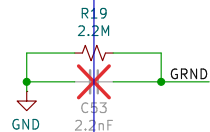
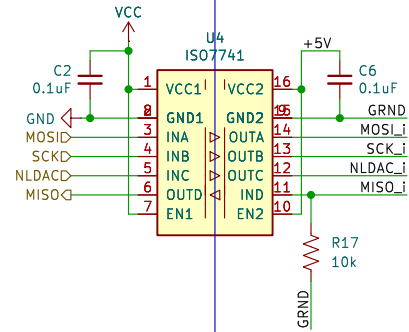
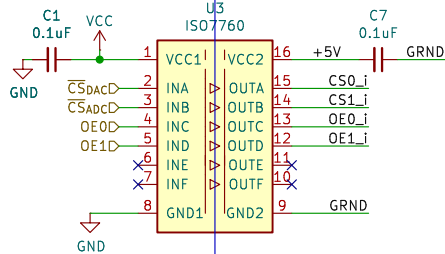
Rev: B

Id: 3/4

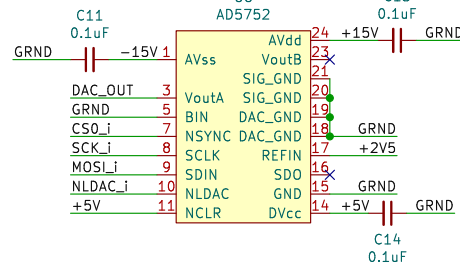
POWER RAILS
power rails are defined as local nets since they are galvanically isolated



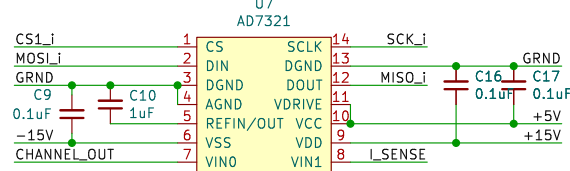
SIGNAL ISOLATION



DAC



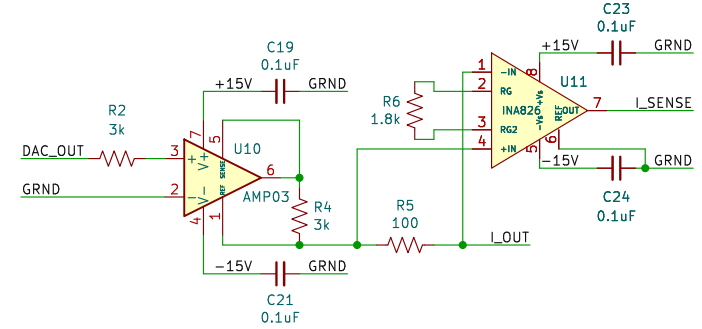
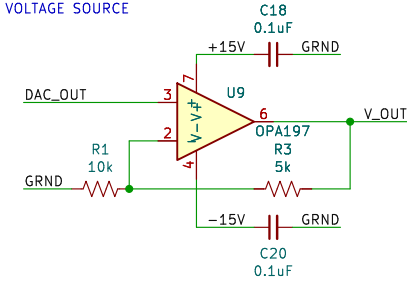
ADC



+15V ○ TP15 TestPoint
-15V ○ TP16 TestPoint
+5V ○ TP19 TestPoint
+2V5 ○ TP21 TestPoint
GND ○ TP42 TestPoint
GND ○ TP43 TestPoint
GND ○ TP44 TestPoint

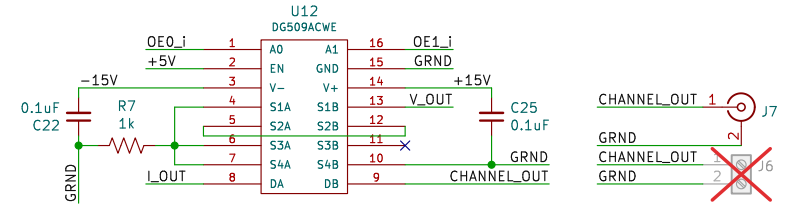
L_OUT ○ TP27 TestPoint
L_SENSE ○ TP29 TestPoint
DAC_OUT ○ TP23 TestPoint
V_OUT ○ TP25 TestPoint
MISO_i ○ TP45 TestPoint
MOSI_i ○ TP48 TestPoint
OE1_i ○ TP49 TestPoint
OE0_i ○ TP50 TestPoint
CS1_i ○ TP51 TestPoint
CS0_i ○ TP52 TestPoint

VOLTAGE SOURCE



Max current output (0ohm load) is $(+/-10V)/3k = +/- 3.33mA$.
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output needs to be within $+/- 10V$
current sensing resolution is $10000 mV / 2^{12} / (2.87mV/uA) = 0.85uA$



OE0	OE1	Output
0	0	VOLTAGE
1	0	CURRENT
0	1	High-Z
1	1	GND

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Jonathan Newman
Nate Cermak

Sheet: /channel_1/
File: channel.kicad_sch

Title: StimJm

Size: A4 Date:
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Rev: B
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